

Exam Date & Time: 26-May-2023 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

**FOURTH SEMESTER B.TECH. (ELECTRONICS AND COMMUNICATION ENGINEERING)
DEGREE EXAMINATIONS - MAY/JUNE 2023
SUBJECT: ECE 2255/ECE_2255 DIGITAL SIGNAL PROCESSING
DIGITAL SIGNAL PROCESSING [ECE 2255]**

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

Missing data may be suitably assumed.

- 1A) Sketch the frequency sampling realization of $M=16$ and $\alpha=0$, linear phase FIR filter which has frequency samples $H\left(\frac{2\pi k}{16}\right) = \begin{cases} 1, & k = 0, 1, 2 \\ 0.5, & k = 3 \\ 0, & k = 4, 5, \dots, 7 \end{cases}$ (5)
- 1B) Consider an FIR filter with lattice coefficients $K_1 = 0.45$, $K_2 = -0.61$, $K_3 = 0.7$. Obtain the impulse response of the filter and sketch its direct form structure. (3)
- 1C) Determine the system function of a causal LTI system, with zeros at $z = 0.5$ and $z = 0.8$, and a complex pair of poles at $z = 1.5 e^{j\frac{\pi}{4}}$. State whether the system is stable and justify your answer, with the help of a pole-zero plot. (2)
- 2A) Develop radix-2 DIF FFT algorithm. Illustrate with signal flow diagram for $N=8$. Highlight the computational advantage of this algorithm. (5)
- 2B) Illustrate with mathematical relations, use of DFT/IDFT in determining the circular convolution between two finite duration sequences. Explain how this is used in determining the response of LTI system to the given input. (3)
- 2C) Consider the finite duration signal $x[n] = n$, $0 \leq n \leq 7$ and 0 elsewhere with 8-point DFT $X[k]$. Using suitable properties of DFT, determine sequence $y[n]$ whose 8-point DFT is $Y[k] = \text{Real part of } |X[k]|$ (2)
- 3A) The specifications of the desired low-pass filter are
 • Passband edge: 4kHz, Stopband edge: 8 kHz
 • Passband ripple: 1 dB, Stopband Attenuation: 40 dB
 • Sampling frequency: 24 kHz
 Determine the order and poles of Butterworth filter required to meet the above filter specification. Use bilinear transformation. (5)
- 3B) (3)

For the filter specification given in Question 3A, determine analog system function $H_a(s)$ and use bilinear transformation to obtain $H(z)$ of Butterworth digital filter.

- 3C) Explain how the Goertzel algorithm exploits the periodicity of the complex phase factor and obtain realization of the system to compute the DFT as a linear convolution. (2)

- 4A) Determine the filter coefficients for a linear phase FIR LPF of length $M=7$. The approximate desired frequency specifications for the filter is

$$H_d(e^{j\omega}) = \begin{cases} e^{-j\omega\alpha}, & 0 \leq |\omega| \leq 0.3\pi \\ 0, & \text{elsewhere} \end{cases} \quad (5)$$

Use suitable window with a minimum stop band attenuation of 50dB.

- 4B) Convert the analog filter to its equivalent digital filter whose system function is given by $H(s) = \frac{s+0.4}{s^2+0.8s+25.16}$ using impulse invariance technique. Assume sampling frequency of 10Hz. (3)

- 4C) Obtain the direct-form II realization for the system $H(z) = \frac{(1-z^{-1}+2z^{-2})}{(1+0.2z^{-1})(1-0.5z^{-1}+0.7z^{-2})}$ (2)

- 5A) Describe with mathematical expressions the Blackman-Tukey method of power spectrum estimation. Describe the spectral leakage and spectral resolution problems occurring in estimation of spectra from finite duration observation of signals. (5)

- 5B) Realize an efficient direct form structure of the linear phase FIR filter whose system function is $H(z) = 0.015 - 0.145z^{-1} + 0.268z^{-2} - 0.268z^{-4} + 0.145z^{-5} - 0.015z^{-6}$ (3)
Determine the corresponding input-output equation.

- 5C) For the filter given in Question 5B, write the equations for the magnitude response and phase response. Is this filter suitable for the design of a lowpass filter? Justify your answer. (2)

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